Derivative Securities FNCE 30007

Tutorial 6

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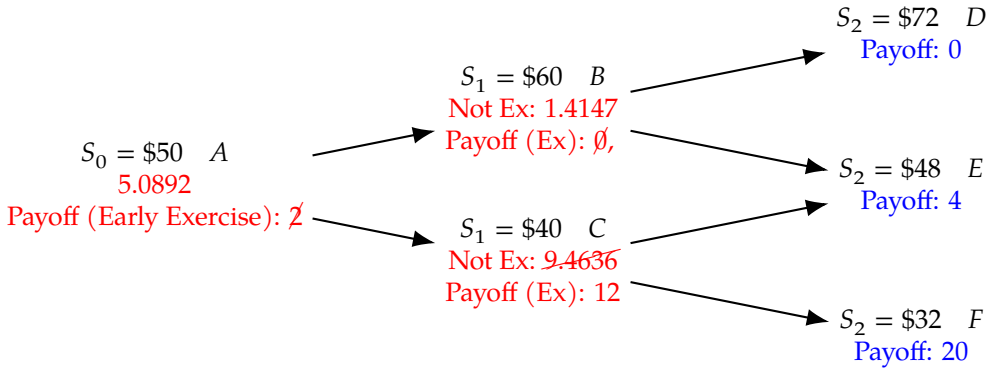
Review

Binomial Model Applied to American Options

- Start at maturity and work backwards thu the stock price tree
- The value at the final nodes is the same as for the Euro option
- At earlier nodes the value of the option is the greater of: (1) the value of an Euro option at that node; (2) the payoff from early exercise.
- Example, $S_0 = 50, K = 52, T_{step} = 1yr, r = 5\%, u = 1.2, d = 0.8$
  $p = \frac{e^{rT} - d}{u - d} = 0.6282, f = (pf_u + (1 - p)f_d)e^{-rT}$

Review

Binomial Model Applied to American Options



Review

Futures Options

- It is an Option, but the underlying asset is the Futures contract. (It's like a derivative of derivative)
- Exercise the call (put) futures option will gives the holder a long (short) position in a futures contract.
  - When the holder exercises the Option at time t, (note t might be different from the option expiry date):
  - Call futures option: Receive cash amount equal to the excess of the futures price over the strike price. $max\{0, F_t - K\}$
  - Put futures option: Receive cash amount equal to the excess of the strike price over the futures price. $max\{0, K - F_t\}$

Review

Futures Options: Put-Call Parity

Portfolio		**Time** 0	**Time** T	
			$F_t \geq K$	$F_t < K$
A	Long Call Futures	c	$F_t - K$	0
	Invest Ke^{-rT} at r	Ke^{-rT}	K	K
B	Long Put Futures	p	0	$K - F_t$
	Long Futures	0	$F_t - F_0$	$F_t - F_0$
	Invest $F_0 e^{-rT}$ at r	$F_0 e^{-rT}$	F_0	F_0
Portfolio Value			A = B	A = B

Table: Put Call Parity

Put-Call Parity: $c + Ke^{-rT} = p + F_0 e^{-rT}$.

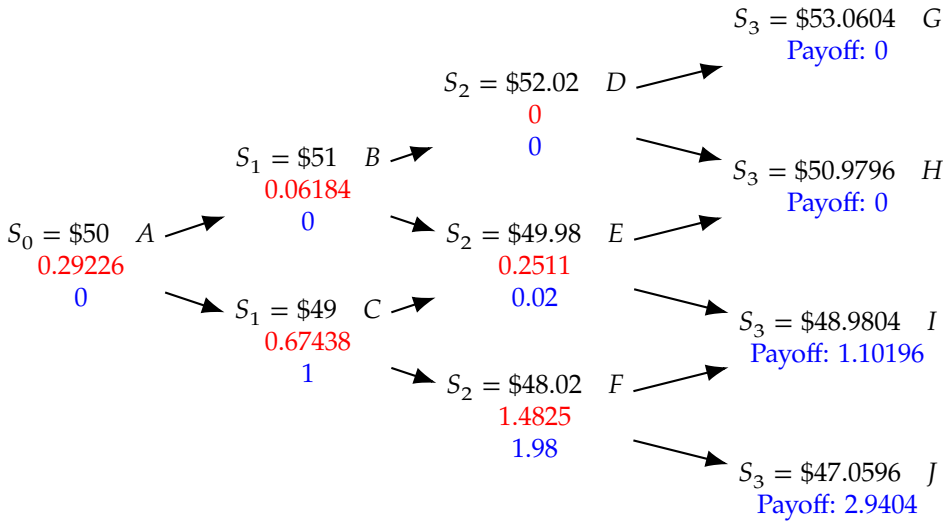
Tutorial Question

Q1: Consider a 3-month American put option on a non-dividend paying stock when the stock price is \$50, the strike price is \$50, the risk-free rate of interest is 12% p.a. and the underlying stock price is expected to increase or decrease by 2% per month.

1. Use a three-period model to calculate the current value of the option.
2. Should the option be exercised early?

- Summarize the information here: $T = 0.25, S_0 = 50, K = 50, r = 12\%, u = 1 + 2\% = 1.02, d = 1 - 2\% = 0.98, \Delta t = 0.25/3 \approx 0.0833$.
- Then, we can find the $e^{r\Delta t} = 1.01005, p = \frac{e^{r\Delta t} - d}{u - d} = 0.75125$

Tutorial Question



Tutorial Question

1. Therefore, the current value of the option is 0.29226.
2. Seemingly, one might choose early exercise at node C or F. But once we are at node C (which means the stock price drops to 49), then we shall exercise since we are not sure whether the next period is at node E or F

Tutorial Question

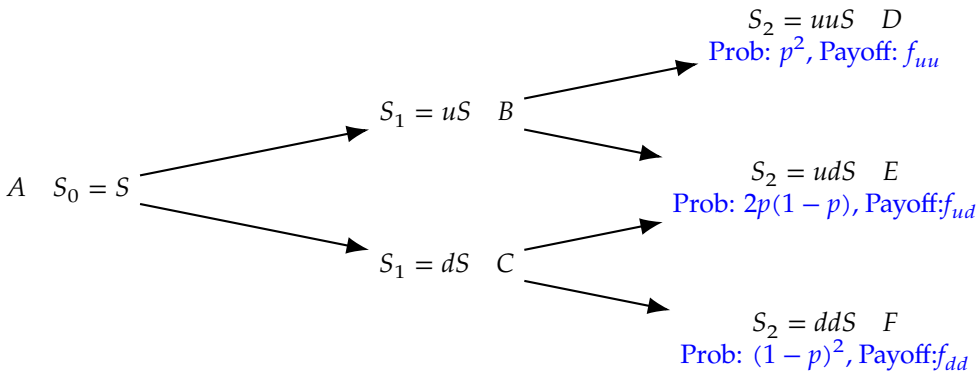
2: Consider a two-period binomial model. Let

$$p = \frac{e^{r\Delta t} - d}{u - d}$$

be the probability of a stock price increase in a risk-neutral world. Use the principle of Risk-Neutral valuation to derive the two-period (European) valuation formula:

$$f = \frac{p^2 f_{uu} + 2p(1-p)f_{ud} + (1-p)^2 f_{dd}}{e^{2r\Delta t}}.$$

Tutorial Question



$$E^*[f_T] = p^2 f_{uu} + 2p(1-p)f_{ud} + (1-p)^2 f_{dd}$$

$$f = \frac{p^2 f_{uu} + 2p(1-p)f_{ud} + (1-p)^2 f_{dd}}{e^{2r\Delta t}}$$

Tutorial Question

3: Using two portfolios, derive the lower bound for the price of a European futures put option. In doing this, detail the composition of each portfolio and draw a table illustrating the initial and terminal values of the portfolios.

Tutorial Question

Portfolio		**Time** 0	**Time** T	
			$F_t \geq K$	$F_t < K$
A	Invest Ke^{-rT} at r	Ke^{-rT}	K	K
B	Long Put Futures	p	0	$K - F_t$
	Long Futures	0	$F_t - F_0$	$F_t - F_0$
	Invest F_0e^{-rT} at r	F_0e^{-rT}	F_0	F_0
Portfolio Value			A $\leq$ B	A = B

Table: Put Lower Bound

From the table above, we find the price lower bound for the put futures option is: $p \geq max\{e^{-rT}(K - F_0), 0\}$

Tutorial Question

Q4: Suppose you buy a put option contract on October gold futures with a strike price of \$900 per ounce. Each contract is for the delivery of 100 ounces. What happens if you exercise when the October futures price is \$880?

An amount $(900 - 880) \times 100 = \$2{,}000$ is added to your margin account, and you acquire a short futures position obligating you to sell 100 ounces of gold in October. This position is marked to market in the usual way until you choose to close it out.

Tutorial Question

Q5: Suppose that a futures price is currently 30. The risk-free interest rate is 5% per annum. A three-month American call futures option with a strike price of 28 is worth 4. Calculate bounds for the price of a three-month American put futures option with a strike price of 28.

The put-call parity for American futures options is: $F_0 e^{-rT} - K < C - P < F_0 - Ke^{-rT}$, bring $C = 4$ and others into the calculation, $1.65 < P < 2.37$